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CAR PARKING SYSTEM(OOP)

MUHAMMAD RAMZAN,
AHTISHAM SALEEM
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Table 1 5

1: GROUP MEMBER:

- MUHAMMAD RAMZAN
  - Contribution:
    Complete project except for detail class and report

- AHTESHAM SALEEM
  - Contribution:
    Detail class and slides.

2: Car parking system:

Car parking system is used to park the cars at some fix position which will be safe from everything. The cars will be parked by some references like the name of driver, number of car, and by knowing the time of departure. Car parking systems have been around almost since the time cars were invented. In any area where there is a significant amount of traffic, there are car parking systems. Car Parking systems were developed in the early 20th century in response to the need for storage space for vehicles. In this project we have made a system of parking the cars by using different functionalities like classes, functions instructors etc.

3: INTRODUCTION OF THE CLASS DIAGRAM:

There we have five classes that are linked with one another through composition, aggregation and inheritance. The class detail is linked through composition with car class and the class calculate is inheritance with the class car. The other two classes are linked with the car class through association.

4: IMPLEMENTATION METHODOLOGY:

1. COMPOSITION:

Composition is generally used when you want the features of an existing class inside your new class, but not its interface. That is, you embed an object to implement features of your new class, but the user of your new class sees the interface you've defined rather than the interface from the original class.

```
class DETAILS
{public:
    car obj;//making composition with car class
    void output()
    {
        cout<<"\n\n\t\tThe Driver Name : "<<obj.dname<<endl;
        cout<<"\n\n\t\tThe Car No : "<<obj.vno<<endl;
        cout<<"\n\n\t\tThe Hours Of Stay : " <<obj.count;
        cout<<"\n\n\t\tThe Time Slot Of Parking : " <<obj.l;

    }
```

Figure 1

2. Association:

An association supports data sharing between classes or, in the case of a self-association, between objects of the same class.

```
class LOGIN;
class DELETE;
class car
{
    LOGIN *ptr;//making association with LOGIN class
    DELETE *ptr2;//making association with DELETE class
    public:
    int vno;
    float count;
    char  dname[15],x,l[50];
```

Figure 2

```
class DELETE{
    public:
        car *ptr;//making association with car class
void delete_record()
```

Figure 3

3. Inheritance:

Inheritance is one of the key features of Object-oriented programming in C++. It allows us to create a new class (derived class) from an existing class (base class). The derived class inherits the features from the base class and can have additional features of its own.

```
class CALCULATE:
    public car
{
    public:
void cal()
```

Figure 4

5. CLASS DIAGRAM:

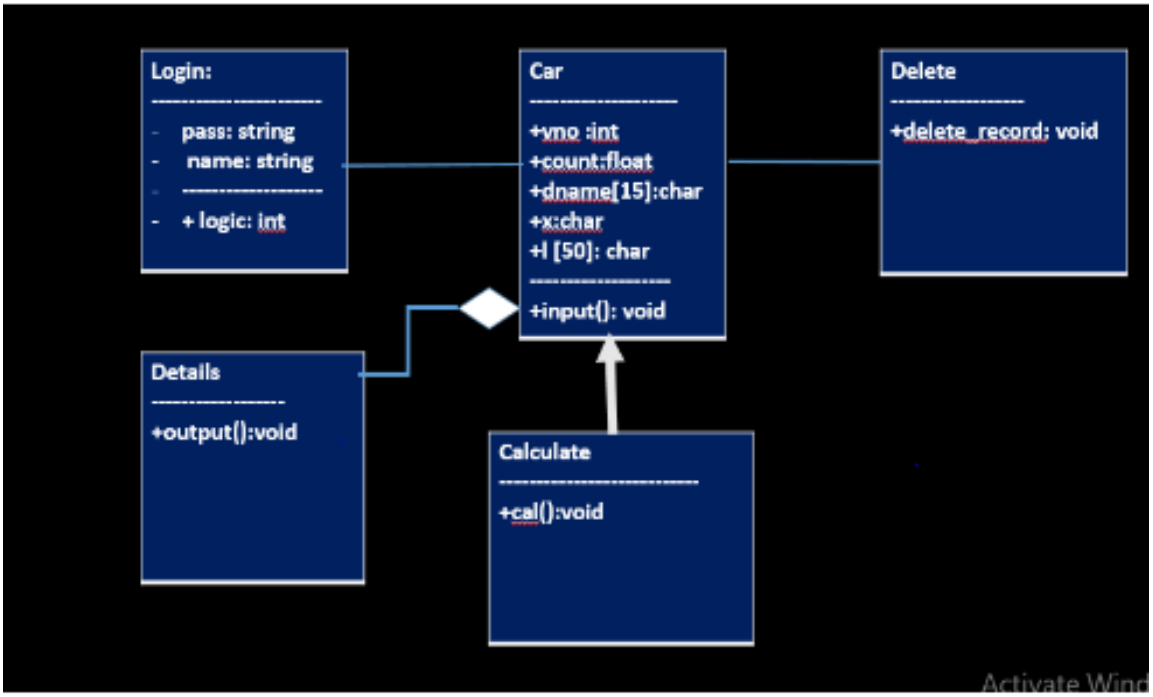

Table 1

6. Outputs:

1: Login system:

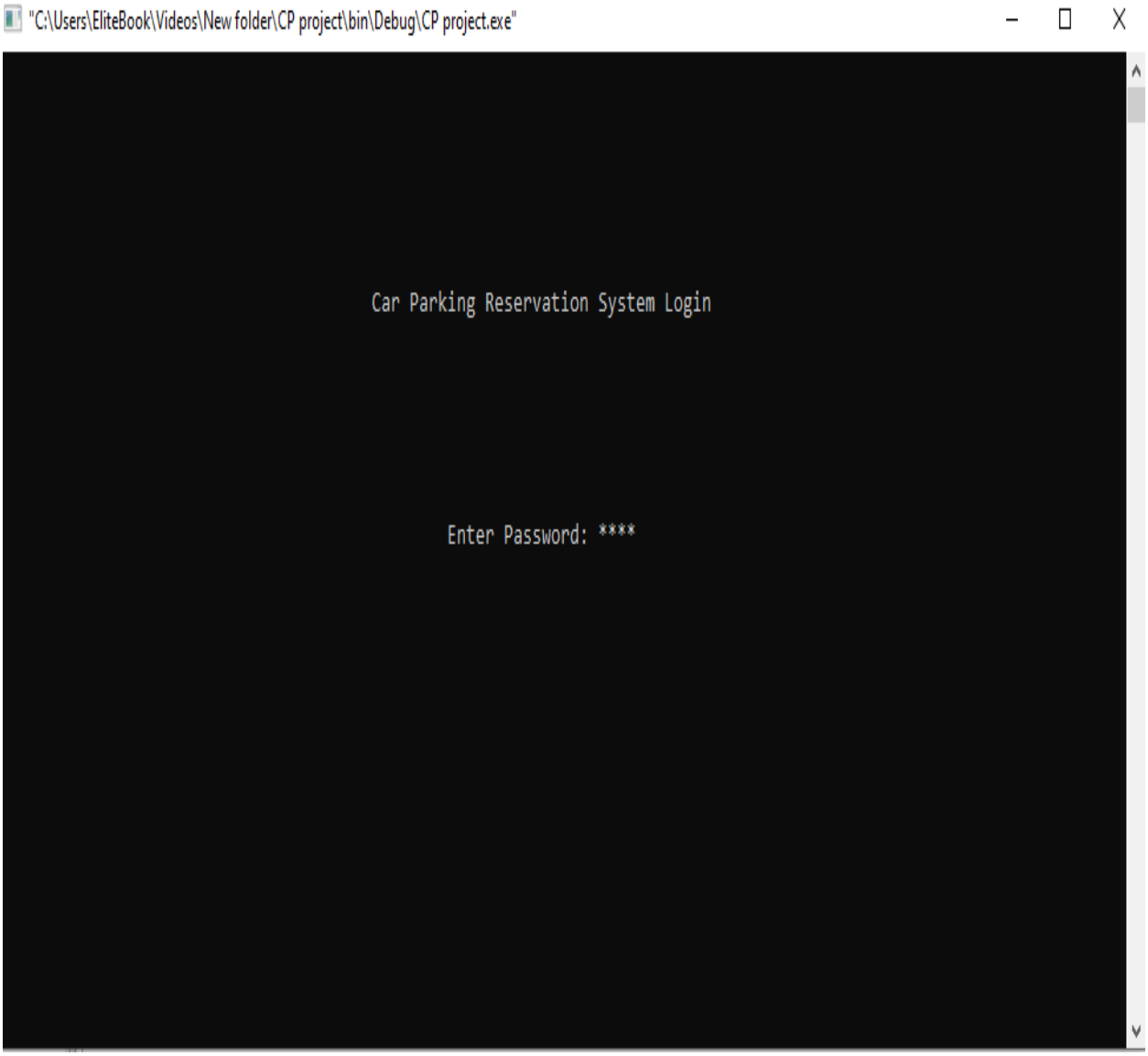
"C:\Users\EliteBook\Videos\New folder\CP project\bin\Debug\CP project.exe"
Car Parking Reservation System Login
Enter Password: ****

Figure 5

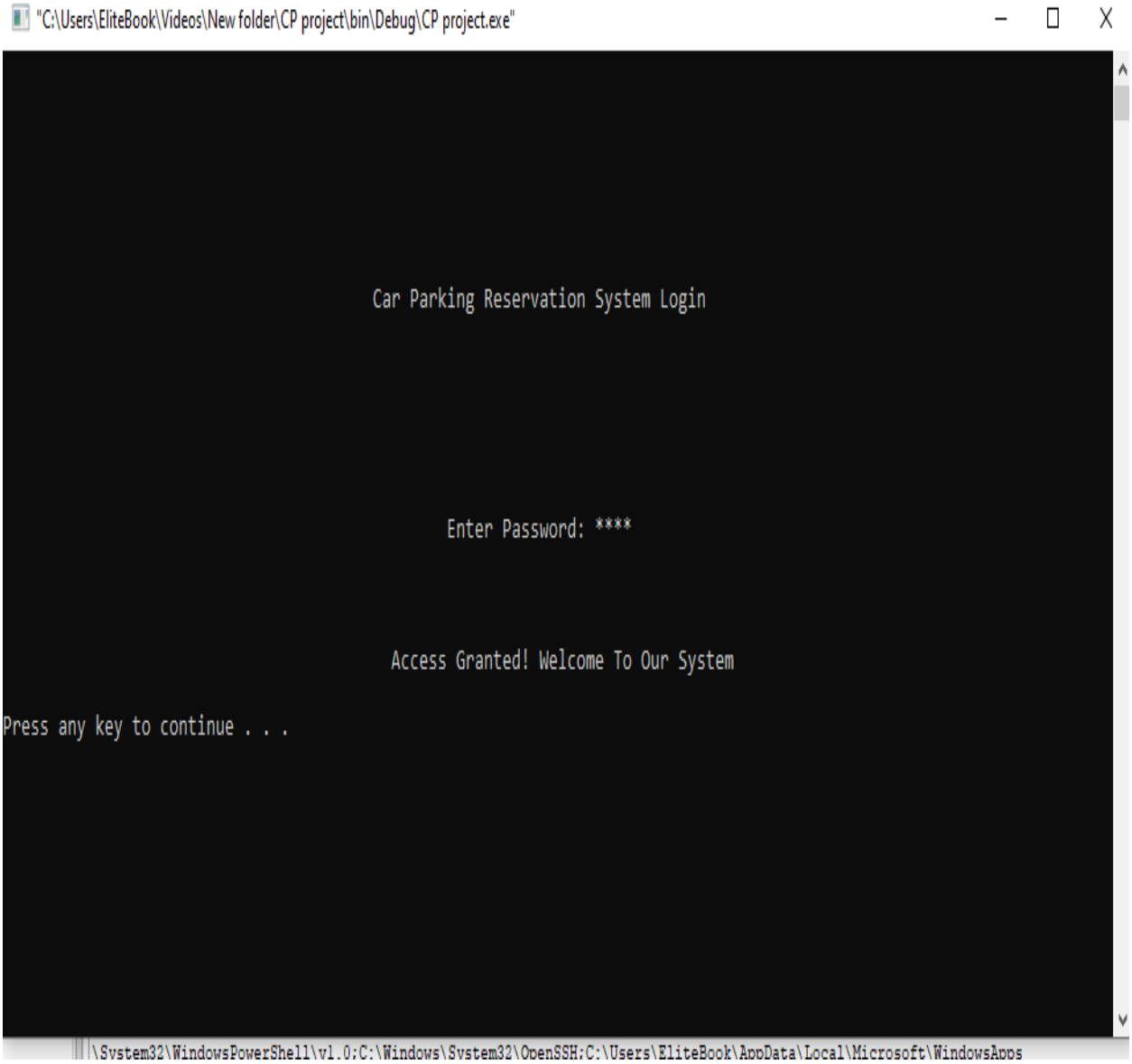
"C:\Users\EliteBook\Videos\New folder\CP project\bin\Debug\CP project.exe"
Car Parking Reservation System Login
Enter Password: ****
Access Granted! Welcome To Our System
Press any key to continue . . .

Figure 6

2: Front page:

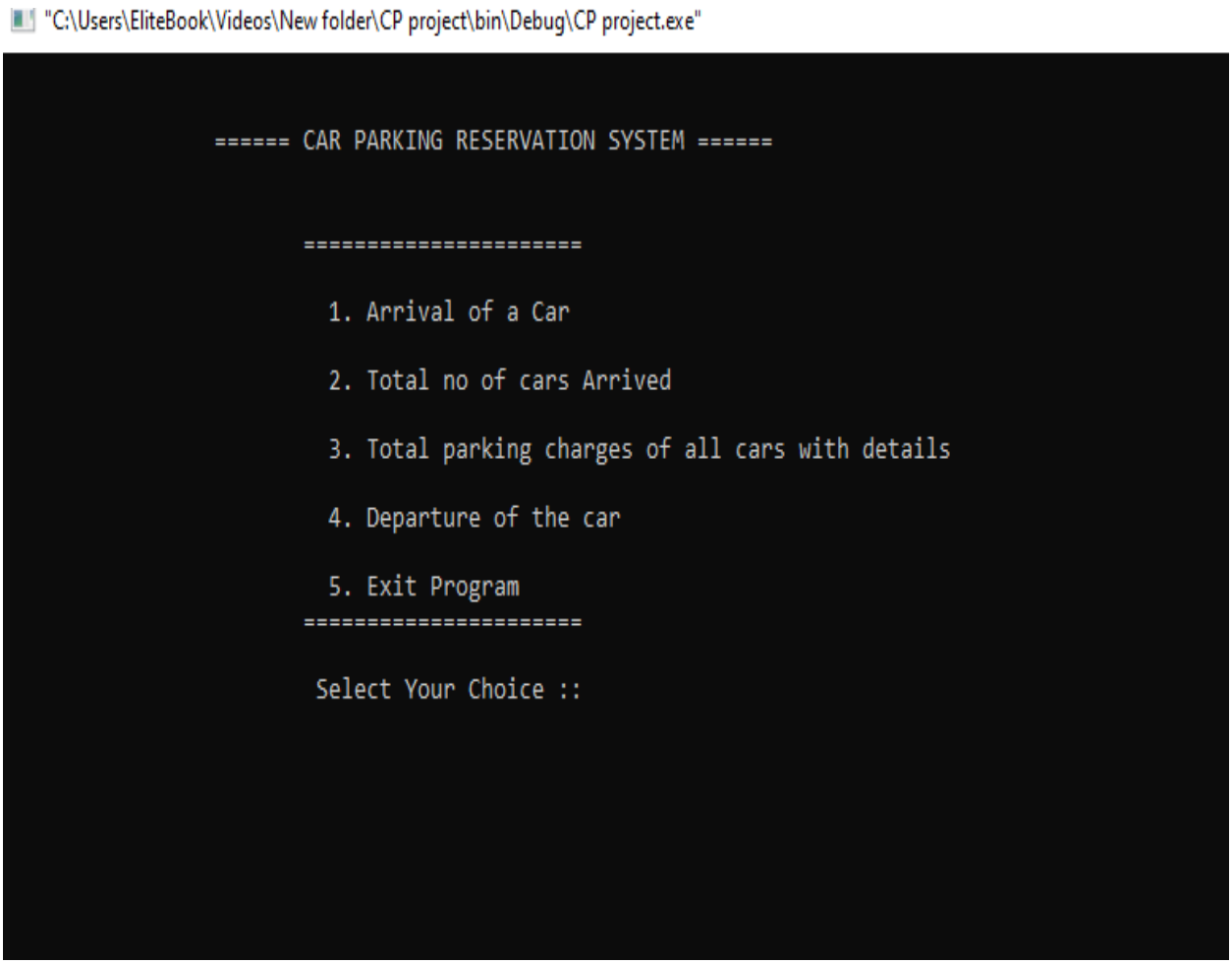

Figure 7

3: Parking:

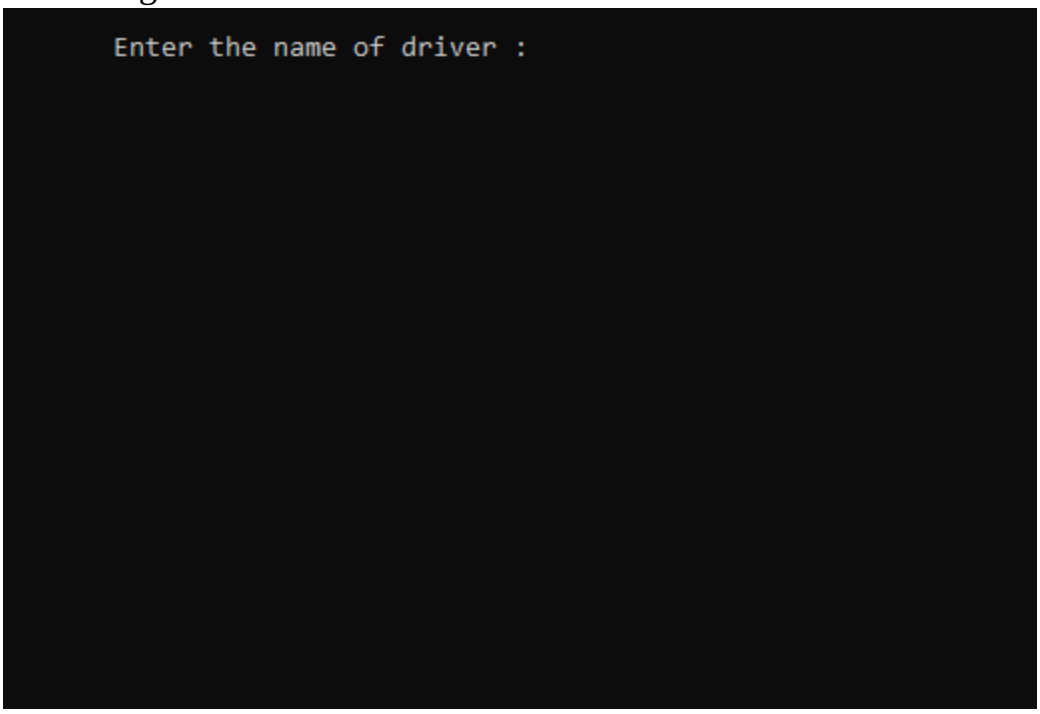

Figure 8

"C:\Users\EliteBook\Videos\New folder\CP project\bin\Debug\CP project.exe"

```
Enter the name of driver : ramzan

Enter the car no : 222

Enter the no of hours of stay : 22

Enter the time slot : 12

Parking Available! You can park your car.

Car got parked!!
```

Figure 9

4: Checking position of car:

```
The Car position in parking lot : 1

        The Driver Name : ramzan

        The Car No : 222

        The Hours Of Stay : 22

        The Time Slot Of Parking : 12

Press any key to continue . . .
```

Figure 10

```
The Car position in parking lot : 1

        The Driver Name : Ç↕@

        The Car No : 8789468

        The Hours Of Stay : 0

        The Time Slot Of Parking : α↔å

ss any key to continue . . .
```

Figure 11

5: Total expenses:

```
The total expense during parking  20 rupees

        The Driver Name : Ç↕@

        The Car No : 8789468

        The Hours Of Stay : 0

        The Time Slot Of Parking : α↔å
```

Figure 12

```
The total expense during parking  1.17682e-037 rupees

        The Driver Name : ramzan


        The Car No : 222


        The Hours Of Stay : 22

        The Time Slot Of Parking : 12



Are you a V.I.P: y/n
```

Figure 13

6: Deleted car:

```
            Enter the car no you want to get depart :
```

Figure 14

```
Enter the car no you want to get depart : 222

 Parking Record Deleted
```

Figure 15

7: Exist program:

```
                    Brought To You By: code-projects.org

Process returned 0 (0x0)   execution time : 219.192 s
Press any key to continue.
```

Figure 16

7. Code:

```
#include<iostream>

#include<bits/stdc++.h>

using namespace std;

#include<conio.h>

#include<string.h>

#include<process.h>

#include<stdio.h>

#include<fstream>

class LOGIN;

class DELETE;

class car

{

   LOGIN *ptr;//making association with LOGIN class

   DELETE *ptr2;//making association with DELETE class

            public:

            int vno;

            float count;

            char  dname[15],x,l[50];

            void input()

            {

               fstream file;

   file.open("Parking.txt",ios::out);

                        system("CLS");

                                    int d;

                        cout<<"\n\n\t\tEnter the name of driver : ";

                        cin>>dname;

                        cout<<"\n\n\t\tEnter the car no : ";

                        cin>>vno;
```

```
                cout<<"\n\n\t\tEnter the no of hours of stay : ";
                cin>>count;
                cout<<"\n\n\t\tEnter the time slot : ";
                cin>>l;
                if(count<50)
                        {
                          file<<endl<<"   "<<dname<<"   "<<vno<<"   "<<count<<"   "<<l;
                                cout<<"\n\n\t\tParking Available! You can park your car. "<<endl;
                                cout<<"\n\n\t\tCar got parked!!";
                        }
                        else
                                {cout<<"\n\n\t\t Parking not possible!!";}



            }
}a,a1;
class DETAILS
{ //public car:
   car obj;
   public:
   void output()
            {
              fstream file;
              file.open("Parking.txt",ios::in);
              while(!file.eof())
      {
         file>>obj.dname>>obj.vno>>obj.count>>obj.l;
      }
                cout<<"\n\n\t\tThe Driver Name : "<<obj.dname<<endl;
```

```
                cout<<"\n\n\t\tThe Car No : "<<obj.vno<<endl;
                cout<<"\n\n\t\tThe Hours Of Stay : " <<obj.count;
                cout<<"\n\n\t\tThe Time Slot Of Parking : " <<obj.l;

        }
};
//brought to you by code-projects.org
class DELETE{
   public:
      car *ptr;//making association with car class
void delete_record()
{
        int n;
        system("CLS");
        cout<<"\n\n\t\tEnter the car no you want to get depart : ";
        cin>>n;
        ifstream inFile;
   inFile.open("parking3.dat", ios::binary);
        ofstream outFile;
   outFile.open("temp.dat", ios::out | ios::binary);
   while(inFile.read((char*)&a, sizeof(a)))
   {
      if(a.vno!= n)
      {

         outFile.write((char*)&a, sizeof(a));
      }

   }
```

```
    cout<<"\n\n\t\t Parking Record Deleted";
    inFile.close();
    outFile.close();
    remove("parking3.dat");
    rename("temp.dat", "parking3.dat");
}
};
class LOGIN
{
    car *ptr;
    string pass ="";
   char name;
public:
    int login();
};
int LOGIN::login(){
   cout <<"\n\n\n\n\n\n\n\t\t\t\t\tCar Parking Reservation System Login";
   cout << "\n\n\n\n\n\n\n\t\t\t\t\t\tEnter Password: ";
   name = _getch();
   while(name != 13){//character 13 is enter
      pass.push_back(name);
      cout << '*';
      name = _getch();
   }
   if(pass == "pass"){
      cout << "\n\n\n\n\t\t\t\t\t  Access Granted! Welcome To Our System \n\n";
      system("PAUSE");

   }else{
```

```
    cout << "\n\n\n\n\t\t\t\t\tAccess Aborted...Please Try Again!!\n";
    system("PAUSE");
    system("CLS");
    login();
  }
}
class CALCULATE:
   public car
{
   public:
void cal()
        {
                cout<<"\n\t\n\tAre you a V.I.P: y/n ";
                cin>>x;
                if(x=='y')
        {
                system("CLS");
                cout<<"\n\t\n\tThe total expense during parking  ";
                cout<<"20 rupees";
        }
        else
        {
                system("CLS");
                cout<<"\n\t\n\tThe total expenses during parking  ";
                cout<<count*25<<" rupees";

        }
}};
int main()
```

```
{
        LOGIN obj;
        CALCULATE obj2;
        DELETE obj3;
        DETAILS obj4;
  obj.login();
        int choice,vno1,d;
        while(1) {
        system("CLS");

   cout << "\n\n\t\t ====== CAR PARKING RESERVATION SYSTEM ======";
   cout <<"\n\n                                    ";
   cout<<" \n\t\t\t======================";
   cout << "\n\n\t\t\t  1. Arrival of a Car";
   cout << "\n\n\t\t\t  2. Total no of cars Arrived";
   cout << "\n\n\t\t\t  3. Total parking charges of all cars with details";
   cout << "\n\n\t\t\t  4. Departure of the car";
   cout << "\n\n\t\t\t  5. Exit Program";
   cout<<" \n\t\t\t======================";
   cout << "\n\n";
   cout << "\t\t\t Select Your Choice ::";
   cin>>choice;
   switch(choice)
   {
    case 1 :
     {
                char f;
                cout<<"\n\n\t\t==Do You Want to continue==y/n ";
                cin>>f;
```

```
                if(f=='y')
                {
                                ofstream f1("parking3.dat",ios::binary|ios::out|ios::app);
        a.input();
        f1.write((char*)&a,sizeof(a));
 }
       break;
 }
case 2 :
   {
       system("CLS");
                    cout << "\n\n\t\t=== View the Records in the Parking Database ===";
   cout << "\n";
   ifstream infile("parking3.dat",ios::binary|ios::in);
   int d=1;
  // while (infile.read((char *)&a,sizeof (a)))
                   // {
                 system("CLS");
   cout << "\n";
   cout<<"\n\n\tThe Car position in parking lot : "<<d;
   d++;
   obj4.output();
   cout<<"\n";
   cout<<"\n\n\n";
   system("PAUSE");
   cout<<endl;
        //}
   break;
        }
```

```
        case 4:
                {
                obj3.delete_record();
                break;
        }
        case 3:
                {
                        ifstream infile("parking3.dat",ios::binary|ios::in);
    //while (infile.read((char *)&a,sizeof (a)))
                //{
      cout << "\n";
                        obj2.cal();
                        obj4.output();
                        cout<<"\n";
                        cout<<endl;

                //}
                break;
}
                case 5:
                {
                        system("CLS");
                        cout<<"\n\n\n\t\t\tBrought To You By: code-projects.org \n";
                exit(0);
                        break;
                }
                default :
                {
                cout<<"\n\n\t\t Invalid input";
```

```
                    cout<<"\n\n\t\tPress Enter to continue";
        }
  }
getch();
}}
//cpo
```

8. CONCLUSION:

This is a program made for parking system. In this program we introduced many features to make parking system more efficient and to ensure real-life implementation. This is a password protected program. This allows a user to first login himself/herself and then enters the details of car to park his car. Then the user can also check the total number of cars parked. Also the user can calculate the total expense or fair of car parked. And at last the user has option to exit the car parked and delete the history of that car. These are all option that we provided the user to use.

Car Parking systems were developed in the early 20th century in response to the need for storage space for vehicles. In this project we have made a system of parking the cars by using different functionalities like classes, functions instructors etc.